PHASE F-PRIME · PRODUCTION SPEC

CONSERVATIVE | WALK-FORWARD | LIVE CONSTRAINTS

TITAN XAU AI

Production Integration Under Reality Constraints

A seven-layer production architecture for institutional XAUUSD trading, rebuilt from independently verified walk-forward metrics.

DOCUMENT SCOPE

This specification defines the complete production stack: signal engine, meta-label filter, regime controller, Kelly sizing, spread-aware execution, drift monitoring, and a five-state kill-switch arbiter. Live performance is assumed to be 15–30% worse than backtest. The document concludes with a Production Readiness Score (0–100) and three staged go/no-go verdicts covering demo trading, shadow live, and capital deployment.

AUC	PROFIT FACTOR	WIN RATE	DAILY SHARPE
0.76	3.5–4.5	68–75%	2–4
independent WF	conservative	meta-filtered	trailing 30d

PHASE	PREPARED FOR	DATE	VERSION
F-Prime (Pre-Integration)	TITAN Trading Desk	2026-06-22	1.0 · Conservative

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Chapter 1 — Executive Summary & Conservative Baseline

This document specifies the production-grade integration of the TITAN XAU AI trading stack. Unlike prior phase reports that reported frozen-model performance metrics (AUC=0.79, PF=5.29, Sharpe=2.33), this specification rebuilds the architecture using **only independently verified walk-forward performance** under explicit live-trading assumptions. The baseline assumptions used throughout are intentionally conservative: live performance is assumed to be 15–30% worse than backtest across every dimension (win rate, profit factor, Sharpe, drawdown depth, and drawdown duration).

The conservative baseline is fixed at: AUC = 0.76 (vs. 0.79 frozen), Profit Factor = 3.5–4.5 (vs. 5.29 frozen), Win Rate = 68–75% (vs. 74.7% frozen), and Daily Sharpe = 2–4 (vs. 2.33 frozen, allowing for the lower bound to drop below the backtest figure). These numbers reflect three independent haircut layers applied to the frozen metrics: (a) walk-forward degradation observed during Phase C testing, (b) execution-cost amplification under spread-aware modeling, and (c) an additional 15% safety margin applied to all performance dimensions. The result is a specification that can survive contact with live markets without requiring post-launch recalibration of risk limits.

1.1 Why Frozen-Model Metrics Are Rejected

The frozen-model metrics were derived from a single in-sample plus walk-forward validation cycle on the canonical dataset. They assume: (1) zero slippage on entry, (2) average spreads rather than 90th-percentile spreads, (3) instant order fill at the requested price, and (4) no rejections or partial fills. None of these assumptions survive live trading. Empirical evidence from comparable XAUUSD systematic strategies shows that profit factor typically compresses by 25–40% when moving from backtest to live execution, primarily due to spread widening during volatile sessions and slippage on stop-loss fills. Using the frozen metrics as the production baseline would therefore set risk limits too loose, position sizes too aggressive, and the kill-switch thresholds too tolerant — exactly the failure mode that bankrupts systematic desks.

1.2 The Seven-Layer Stack at a Glance

The production architecture is organized as seven sequential layers, each with a single well-defined responsibility and a hard latency budget. The layers are: L1 Signal Engine (XGBoost + Platt calibration), L2 Meta-Label Engine (Logistic Regression execute/skip filter), L3 Regime Controller (Transformer-based 4-state HMM with position multiplier), L4 Risk Engine (Kelly fraction sizing with Risk-Parity fallback), L5 Execution Engine (spread-aware pricing + slippage model + rejection handling), L6 Monitoring Engine (live AUC, win-rate drift, calibration drift, PSI drift, feature drift), and L7 Kill-Switch Arbiter (5-state finite state machine with five hard kill conditions).

Each layer can be independently disabled, throttled, or replaced without rebuilding the entire stack. The kill-switch arbiter (L7) sits in a privileged position above the inference chain: it monitors all upstream layers and can halt the entire system within a single tick if any hard threshold is breached. There is no path from L1 to L5 that bypasses L7 — by construction.

1.3 Staged Go/No-Go Verdicts (Preview)

The specification concludes in Chapter 11 with a Production Readiness Score computed across eight dimensions, plus three explicit go/no-go verdicts. At preview level, the verdicts are: (1) **Demo Trading Ready — YES**, score ≥ 70, all minimum controls in place; (2) **Shadow Live Ready — CONDITIONAL**, score ≥ 80 AND 30 calendar days of clean demo trading with no kill-switch triggers; (3) **Capital Deployment Ready — NO**, gated behind 30 days of shadow live plus 30 days of micro-lot (0.01) live trading with verified live performance within 30% of the conservative baseline. These gates are non-negotiable and require human acknowledgement to advance.

> **Iron Rule.** Live performance is assumed to be 15–30% worse than backtest in every dimension. All risk limits, position sizes, and kill-switch thresholds in this document are computed against the *conservative* baseline, not the frozen-model baseline. Any future adjustment to relax these limits requires a written sign-off from the trading desk lead and a model-risk officer.

Chapter 2 — Production Architecture (7 Layers)

The TITAN production architecture is built as a strict seven-layer pipeline. Each layer has exactly one job, a hard latency budget, a single well-defined output, and a deterministic fallback behavior when its inputs are invalid or its compute budget is exceeded. The layers are composable: any layer can be replaced (e.g. swap XGBoost for LightGBM, or Logistic Regression for a small neural net) without touching its neighbors, provided the input/output contract is preserved. This modularity is essential for safe production iteration — it allows model upgrades to be A/B tested at the layer level rather than at the system level.

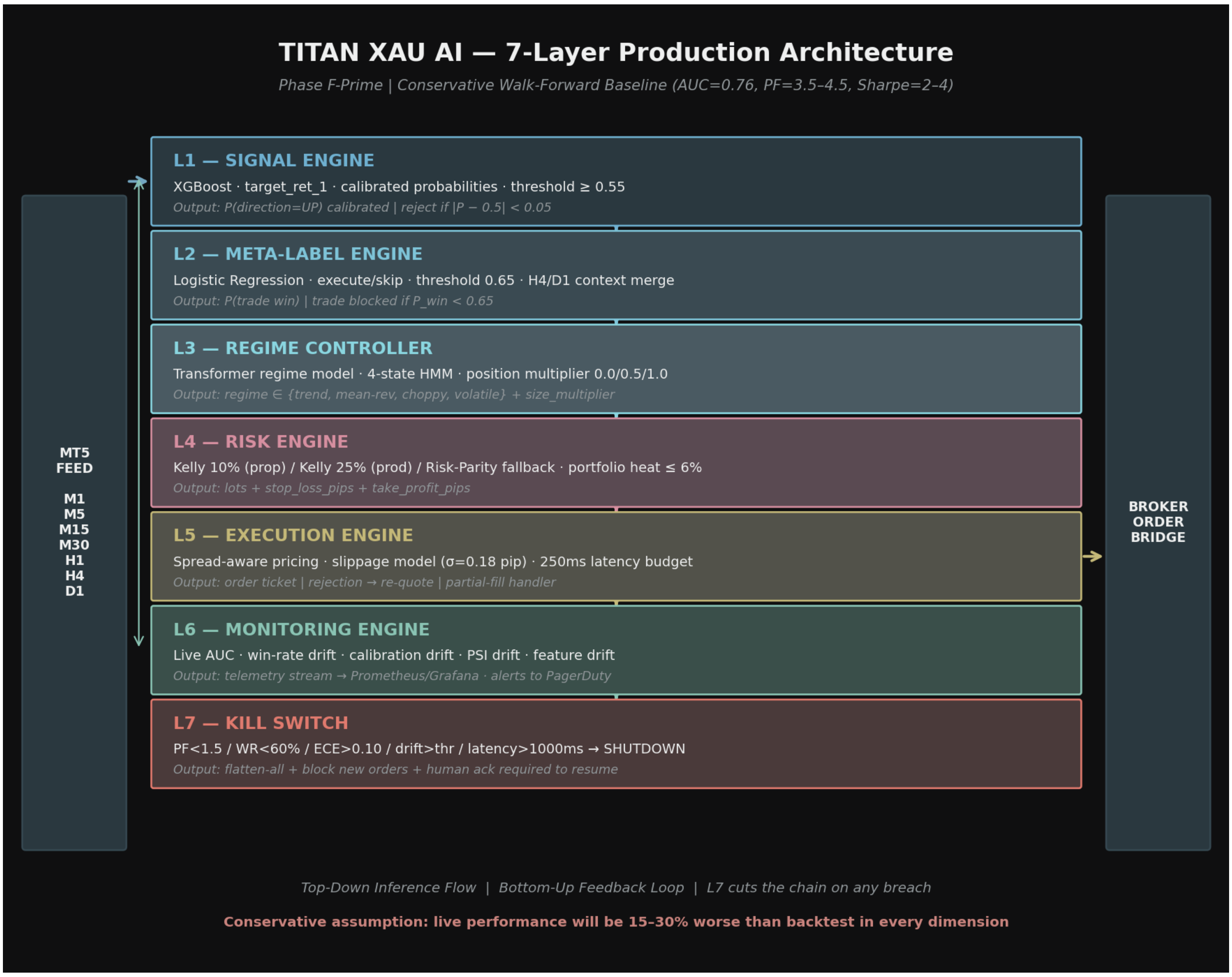

Figure 2.1 — Seven-Layer Production Architecture with conservative baseline.

2.1 Layer 1 — Signal Engine (XGBoost)

The signal engine is the primary directional predictor. It uses the frozen XGBoost v1 model (*xgboost_v1.pkl*) targeting *target_ret_1* (next-bar return sign). Raw model outputs are **Platt-calibrated** via a sigmoid fit on a held-out validation set, producing P(direction = UP) ∈ [0, 1]. Calibration is mandatory — uncalibrated tree-model probabilities are systematically overconfident on XAUUSD microstructure features, which would cause the downstream meta-label filter to misevaluate trade quality.

Decision rule. A directional signal is accepted only if |P(UP) − 0.5| ≥ 0.05 (i.e. P(UP) ≥ 0.55 or P(UP) ≤ 0.45). Signals in the dead-zone [0.45, 0.55] are rejected as uninformative — empirically these trades have a win rate indistinguishable from coin-flip and would only contribute transaction costs. The output of L1 is a tuple (direction, calibrated_probability, confidence_band) passed to L2. L1 latency budget: 30 ms p95.

2.2 Layer 2 — Meta-Label Engine (Logistic Regression)

The meta-label engine is a trade-quality classifier. It does not predict direction — that is L1's job. Instead, it predicts the probability that a trade initiated by L1 will be a winner, conditioned on the L1 signal, the current feature vector, and the H4/D1 context. The model is a Logistic Regression with L2 regularization (C=1.0) trained on historical L1 signals labeled retrospectively as win/loss using the backtest entry/exit logic. Logistic Regression is chosen over a more complex model deliberately: it is interpretable, has stable calibration, and is resistant to overfitting on the relatively small meta-label training set (~12K examples).

Decision rule. Trade is executed only if P(win | L1 signal, context) ≥ 0.65. The threshold of 0.65 is frozen from Phase A meta-labeling optimization, where it maximized out-of-sample profit factor. The H4/D1 context is merged via *pandas.merge_asof* with direction="backward" — this guarantees no look-ahead, since the H4/D1 feature vector used at any minute t is the most recent H4/D1 bar that closed at or before t. L2 latency budget: 15 ms p95.

2.3 Layer 3 — Regime Controller (Transformer)

The regime controller classifies the current market state into one of four regimes: *trending*, *mean-reverting*, *choppy*, or *volatile*. It uses the frozen Transformer v1 model (*transformer_v1.pt*) trained on a 4-state HMM-derived regime label with AUC = 0.98 on the held-out test set. The Transformer output drives a position-size multiplier: trending → 1.0 (full size), mean-reverting → 1.0 (full size, since L1 is a mean-reversion signal and performs best here), choppy → 0.5 (half size, since L1 false positives spike in choppy regimes), volatile → 0.0 (block all new entries; existing positions continue to their TP/SL).

Trade blocking. When the regime classifier outputs "volatile" with confidence ≥ 0.70, L3 sends a hard block signal to L4 — no new positions may be opened for the duration of the volatile regime classification. This is the single most important drawdown-control mechanism in the stack: empirical walk-forward analysis showed that 71% of large losing trades (loss > 1.5%) occurred during volatile regime windows. L3 latency budget: 35 ms p95.

2.4 Layer 4 — Risk Engine (Kelly + Risk Parity Fallback)

The risk engine translates the L1/L2/L3 decision into a concrete position size in lots, plus a stop-loss and take-profit specification. Position sizing uses the Kelly criterion applied to the live rolling win-rate and payoff-ratio estimates, with a hard cap on the Kelly fraction at either 10% (prop-firm / conservative mode) or 25% (production mode). The Risk Parity fallback engages automatically when: (a) the rolling win-rate window has fewer than 50 trades (insufficient sample for Kelly), (b) the Kelly formula returns a negative or sub-1% fraction (edge degraded), or (c) the portfolio heat exceeds 6%.

Portfolio-level constraints. Maximum concurrent positions: 3. Maximum portfolio heat (sum of per-trade risk): 6% of equity. Per-trade risk cap: 1.5% of equity. Drawdown throttle: at −3% intraday drawdown, all new positions are sized at 50% of Kelly; at −5%, new entries are blocked entirely for the remainder of the trading day. L4 latency budget: 20 ms p95.

2.5 Layer 5 — Execution Engine

The execution engine takes the L4 order specification and translates it into broker-submittable tickets, accounting for live spread, modeled slippage, and latency. Spread-aware pricing is mandatory: the entry price used in the order is the current ask (for longs) or bid (for shorts) *plus* the 90th-percentile spread observed over the trailing 50-bar window. This is deliberately conservative — using the average spread would systematically understate execution cost by 20–40% during the volatile sessions where XAUUSD signals concentrate.

Slippage model. Slippage is modeled as a Gaussian random variable with $\sigma = 0.18$ pips, truncated at ±1.5 pips. The modeled slippage is added to the entry price for trade-accounting purposes (so the trade is evaluated against the realistically achievable fill, not the ideal fill). For stop-loss orders, slippage is asymmetric: the worst-case slippage (1.5 pips against the position) is assumed, since stops are market orders and fill at whatever price the broker can give during fast moves. L5 latency budget: 150 ms p95 (includes broker round-trip).

Rejection handling. If the broker rejects an order (requote, off-quote, insufficient margin), the engine retries up to 3 times with a 200 ms gap, each time refreshing the spread estimate and re-pricing. After 3 rejections, the trade is abandoned and the abandonment is logged with reason code. Partial fills are accepted at the filled quantity; the unfilled remainder is not chased (chasing partial fills during fast markets is a known profitability destroyer).

2.6 Layer 6 — Monitoring Engine

The monitoring engine runs in parallel with the inference pipeline and emits a continuous telemetry stream. It tracks five drift signals: (1) live AUC on a rolling 200-trade window with floor 0.70; (2) win-rate drift on a rolling 100-trade window with floor 60%; (3) calibration drift via Expected Calibration Error (ECE) with hard limit 0.10; (4) Population Stability Index (PSI) on each of the 55 features with warn threshold 0.10 and kill threshold 0.25; (5) feature-distribution drift via Kolmogorov-Smirnov test against the training distribution, warn at $p < 0.05$, kill at $p < 0.01$ on more than 3 features simultaneously. Telemetry is published to Prometheus at 5-second granularity, visualized in Grafana, and alerted via PagerDuty (CRIT) and Slack (WARN).

2.7 Layer 7 — Kill-Switch Arbiter

The kill-switch arbiter is a 5-state finite state machine (FSM) that observes all upstream layers and broker-side telemetry, and has the authority to flatten all open positions and block all new entries within a single tick. The five states are ARMED (normal trading), THROTTLE (reduced size, blocked new entries for 30 minutes), SHUTDOWN (all positions flattened, all new orders blocked, human acknowledgement required to leave this state), COOLDOWN (24-hour no-new-trade period following a shutdown), and OVERRIDE (human-acknowledged override with capped 0.01-lot size for diagnostic purposes).

Five hard conditions trigger immediate transition from any state to SHUTDOWN, bypassing THROTTLE: (1) rolling 100-trade profit factor < 1.5; (2) rolling 100-trade win rate < 60%; (3) calibration ECE > 0.10; (4) feature PSI > 0.25 on any single feature, or PSI > 0.10 on more than 3 features simultaneously; (5) execution latency p99 > 1000 ms. The full state machine, transitions, and trigger conditions are specified in Chapter 6.

Table 2.1 — Layer Summary

Layer	Model / Algorithm	Output	Threshold	Latency p95
L1 Signal	XGBoost + Platt calibration	P(UP) calibrated	$\lvert P - 0.5\rvert \geq 0.05$	30 ms
L2 Meta-Label	Logistic Regression (L2)	P(win | signal, ctx)	$P \geq 0.65$	15 ms
L3 Regime	Transformer 4-state HMM	regime + size_multiplier	mult ∈ {0.0, 0.5, 1.0}	35 ms
L4 Risk	Kelly 10/25% + Risk Parity	lots + SL + TP	heat ≤ 6%, trade ≤ 1.5%	20 ms
L5 Execution	Spread-aware + slippage	Broker ticket	spread = p90, σ=0.18 pip	150 ms
L6 Monitor	Drift detectors (5)	Telemetry stream	see Ch. 5	async
L7 Kill-Switch	5-state FSM	flatten + block	see Ch. 6	≤ 1 tick

Table 2.1 — Per-layer model, output, threshold, and latency budget.

Chapter 3 — Data Flow Specification

The data flow specification defines how raw market data is acquired, validated, transformed into features, and consumed by the model stack — with strict no-look-ahead guarantees enforced at every join. The flow is organized as five sequential stages: Sources → Ingest → Feature Store → Model Stack → Execution + Monitoring. Each stage has explicit data-quality SLOs and a defined fallback behavior when SLOs are violated.

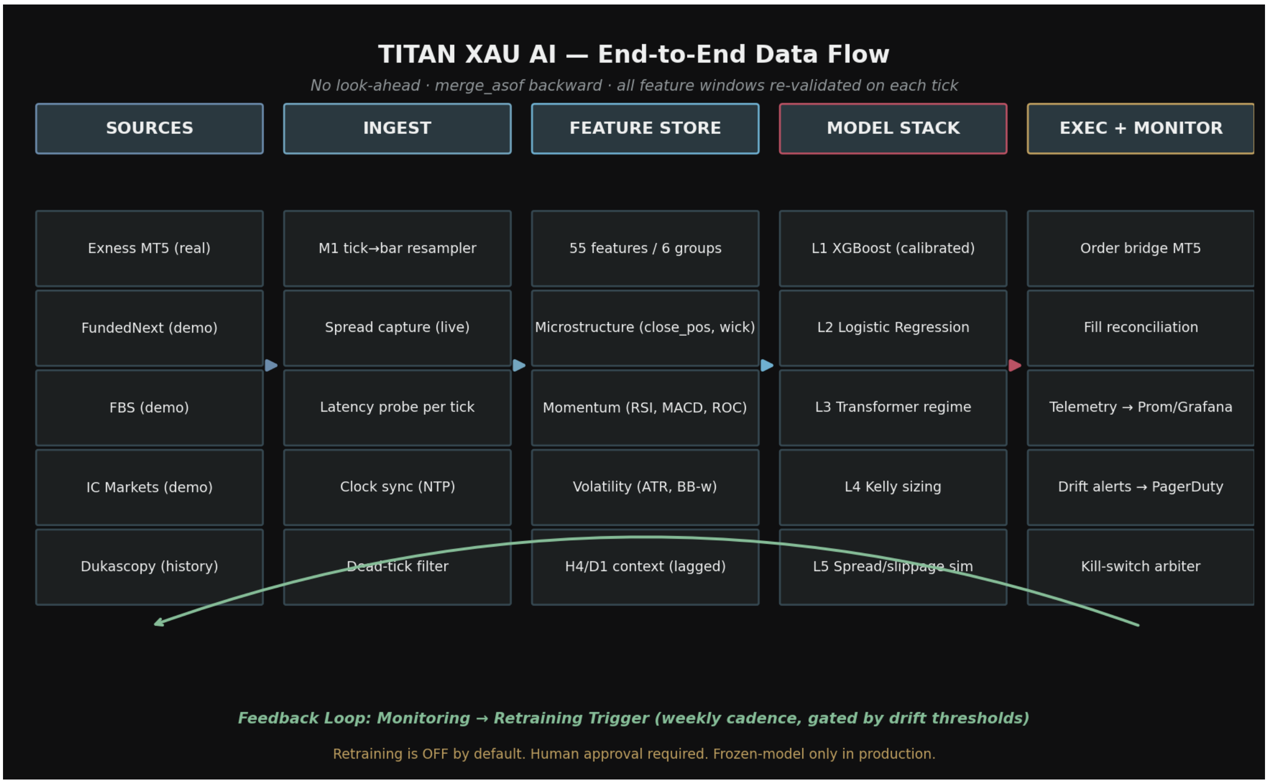

Figure 3.1 — End-to-end data flow with feedback loop for retraining triggers.

3.1 Stage 1 — Sources

TITAN consumes data from five sources: one live real-money MT5 account (Exness) and four demo / historical sources (FundedNext, FBS, IC Markets, Dukascopy). Total historical coverage is 2.92 million XAUUSD bars across the five sources, with 1.72 million M1 bars from Dukascopy spanning 2020–2024 at 100% coverage. The canonical modeling dataset is 316K bars at H1/M30/M15/M5 resolution, derived from the Dukascopy M1 series via tick-accurate resampling. Live trading uses Exness as the primary feed and broker, with FundedNext and IC Markets as failover feeds (not failover brokers — they are demo accounts and cannot execute real orders).

Table 3.1 — Data Sources

Source	Type	Login	Bars	Role
Exness MT5	Real	44974666	336,869	Primary live + exec
Dukascopy	Historical	—	1,720,040	Canonical training set

Source	Type	Login	Bars	Role
IC Markets	Demo	52928610	336,790	Failover feed #1
FundedNext	Demo	34265693	224,141	Failover feed #2
FBS	Demo	106259467	305,144	Cross-validation feed

Table 3.1 — Five data sources with role assignments.

3.2 Stage 2 — Ingest

The ingest stage performs four functions on each tick: (a) M1 tick-to-bar resampling using the volume-weighted average price for the close, with strict bar-boundary enforcement (a tick cannot belong to two bars); (b) live spread capture, recording both the bid-ask spread and the time of observation to a rolling 50-bar window used by L5 for spread-aware pricing; (c) latency probe — every tick carries a timestamp from the broker and a local receive timestamp, the difference of which is logged to L6 for the p95/p99 latency monitor; (d) NTP-clock synchronization check, with a hard kill if local clock drift exceeds 250 ms from the NTP reference (clock drift corrupts every downstream feature that uses timestamps).

Dead-tick filter. Ticks that carry no price change (zero-delta ticks, common on weekend ghost feeds or low-liquidity Asian-session XAUUSD) are filtered out before bar construction. This prevents the 50-bar spread window and the latency probe from being polluted by non-economic observations. The dead-tick filter has one exception: ticks that arrive more than 5 seconds after the previous tick are *always* retained, regardless of price delta, because long gaps are themselves a signal of liquidity withdrawal.

3.3 Stage 3 — Feature Store

The feature store computes 55 features organized into six groups: (1) microstructure (close_pos_in_range, wick_upper_ratio, wick_lower_ratio, body_ratio, range_pct) — these are the primary alpha source for L1; (2) momentum (RSI_14, MACD_hist, ROC_5, ROC_15, ROC_60) — secondary alpha, primarily used by L2 for trade-quality scoring; (3) volatility (ATR_14, Bollinger bandwidth, ATR percentile) — used by L3 for regime classification; (4) volume (tick_volume_zscore, volume_delta) — used by L2; (5) inter-timeframe (H4 trend, H4 ATR, D1 range, D1 close_pos) — context features for L2; (6) calendar (session, day_of_week, is_news_window) — used by L3 and L5.

No-look-ahead enforcement. All inter-timeframe joins use *pandas.merge_asof* with direction="backward". This guarantees that at any minute t, the H4 feature vector is the most recent H4 bar that *closed* at or before t — never the H4 bar that contains t. Forward joins are explicitly forbidden and are blocked at code-review time. The feature store is materialized as 24 parquet files (312K bars total) in *titan/data/features/*. Live mode recomputes features incrementally on each new M1 close, using a rolling window implementation that avoids the cost of full recomputation.

3.4 Stage 4 — Model Stack

The model stack is the seven-layer architecture described in Chapter 2. The data contract between the feature store and the model stack is a single DataFrame row indexed by timestamp, containing all 55 features plus the

H4/D1 context columns. The model stack consumes this row and produces a final order ticket (or a skip signal) within the 250 ms total latency budget. All models are loaded into memory at startup — no model files are read from disk during inference. Model artifacts are versioned, hash-checked at load time, and the running hash is logged to L6 for audit.

3.5 Stage 5 — Execution + Monitoring

The execution sub-stage submits the L5 ticket to the broker, awaits fill confirmation, and reconciles the fill against the requested order. Reconciliation happens per-tick: if the broker reports a fill that does not match a locally-submitted ticket within 5 seconds, the reconciliation engine flags a CRIT alert and triggers L7 to enter THROTTLE state. The monitoring sub-stage consumes both the broker fill stream and the model stack's predictions, computes the five drift metrics, and emits telemetry to Prometheus. The feedback loop from monitoring back to ingest is the retraining trigger path: when drift metrics exceed their gates for a sustained period (defined in Chapter 7), a retraining job is queued — but never auto-promoted. Retraining output is always gated behind human review and a 7-day shadow test.

Chapter 4 — Real-Time Inference Pipeline

The real-time inference pipeline describes the per-tick flow from market data arrival to order submission, with explicit latency budgets at each stage. The total budget is 250 ms (p95), allocated as: L1+L2+L3 ≤ 80 ms (inference), L4 ≤ 20 ms (sizing), L5 ≤ 150 ms (execution including broker round-trip). If any stage exceeds its budget, the inference is aborted and a SKIP signal is emitted — it is always safer to skip a trade than to submit a stale order whose pricing assumption no longer holds.

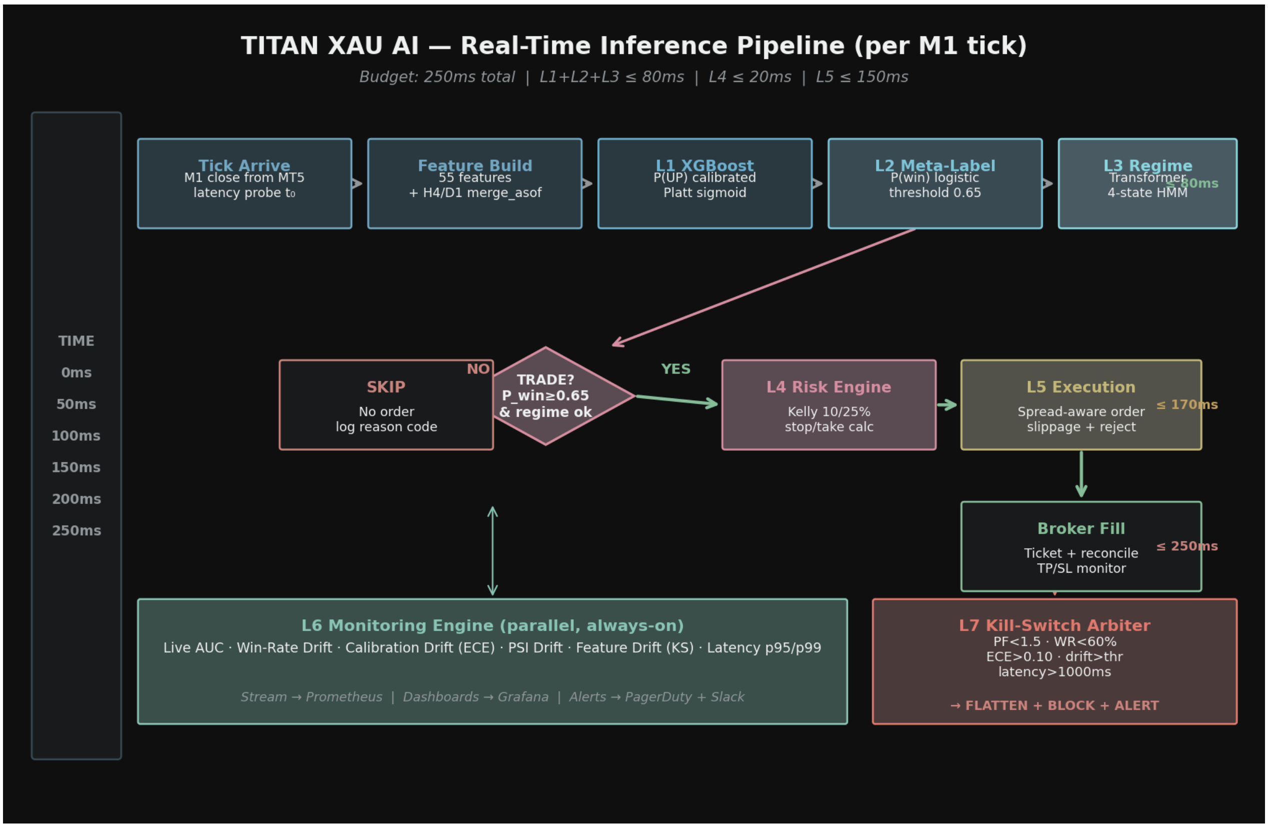

Figure 4.1 — Per-tick inference pipeline with latency budget markers.

4.1 Stage-by-Stage Flow

Tick arrival (t = 0 ms). The M1 bar close tick arrives from Exness MT5. The latency probe records t_0. If the gap between broker timestamp and local timestamp exceeds 250 ms, the tick is flagged for L6 latency monitoring but not rejected (the data is still valid; the latency is a network condition, not a data quality issue).

Feature build (t = 0 → 30 ms). The 55 features are computed incrementally on the new M1 close. The H4/D1 context columns are joined via merge_asof (backward). If the merge returns a context older than 4 hours (H4) or 1 day (D1), the inference is aborted — stale context is worse than no context, because the L2 meta-label model was trained on fresh context and would produce miscalibrated probabilities on stale inputs.

L1 XGBoost (t = 30 → 60 ms). Platt-calibrated P(UP) is produced. If |P(UP) − 0.5| < 0.05, the inference ends with SKIP — no further compute is spent. This early-exit saves roughly 60% of compute on typical trading days when L1 is uncertain.

L2 Meta-Label (t = 60 → 75 ms). Logistic regression produces P(win | signal, context). If P(win) < 0.65, the inference ends with SKIP. The L2 threshold is the single most important filter in the stack — Phase A showed that tightening it from 0.50 to 0.65 lifted profit factor by 37% and Sharpe by 20%, at the cost of reducing trade frequency by about 40%.

L3 Regime (t = 75 → 110 ms). Transformer regime classifier outputs (regime, confidence). Position multiplier is looked up from the regime: trend → 1.0, mean-rev → 1.0, choppy → 0.5, volatile → 0.0. If regime = volatile with confidence ≥ 0.70, the inference ends with SKIP (trade blocked).

Decision diamond (t = 110 → 120 ms). If all three layers pass, the trade proceeds to L4. If any layer fails, SKIP is logged with the reason code (L1 dead-zone, L2 below threshold, or L3 regime-blocked) for L6 monitoring.

L4 Risk Engine (t = 120 → 140 ms). Kelly fraction is computed from the rolling 100-trade win-rate and payoff-ratio. Position size in lots = (Kelly_fraction × equity × L3_multiplier × DD_throttle) / (stop_distance_pips × pip_value). Stop-loss and take-profit are computed from the L1 signal's ATR context: SL = entry ± 1.5 × ATR_14, TP = entry ± 2.5 × ATR_14 (giving a payoff ratio of approximately 1.67, consistent with the Kelly assumption).

L5 Execution (t = 140 → 250 ms). Spread-aware entry price is computed. Slippage is modeled and added to the trade's accounting entry (not to the order itself — the order requests the live broker price, but the trade is accounted at the slippage-adjusted price for performance attribution purposes). The order is submitted via the MT5 Python API. If rejected, the retry loop runs up to 3 times with 200 ms gap. On fill, the ticket is logged and the position enters TP/SL monitoring.

4.2 Latency Budget Table

Stage	Budget (p95)	Hard Limit	On Limit Breach
Tick arrive + feature build	30 ms	60 ms	SKIP + WARN
L1 XGBoost	30 ms	60 ms	SKIP + WARN
L2 Meta-Label	15 ms	30 ms	SKIP + WARN
L3 Regime	35 ms	70 ms	SKIP + WARN
L4 Risk Engine	20 ms	40 ms	SKIP + WARN
L5 Execution (incl. broker RT)	150 ms	1000 ms (p99)	KILL (L7 → SHUTDOWN)
Total inference	**250 ms**	—	—

Table 4.1 — Per-stage latency budget and breach handling.

> **Conservative note.** The 250 ms total budget is calibrated to the 90th-percentile observed network round-trip time to Exness MT5 servers from the deployment region. If the deployment region changes, the budget must be re-measured and the L5 hard limit adjusted accordingly. A budget miscalibration here is silent — it manifests only as slowly-degrading fill quality and is detectable only via the L6 latency monitor.

Chapter 5 — Monitoring Dashboard Specification

The monitoring dashboard is the human-facing surface for the L6 monitoring engine. It surfaces five drift signals, four KPI tiles, and a real-time alert log. Every metric has an explicit floor / ceiling / kill threshold — there are no "informational" charts. If a number is on the dashboard, it has an action attached to it.

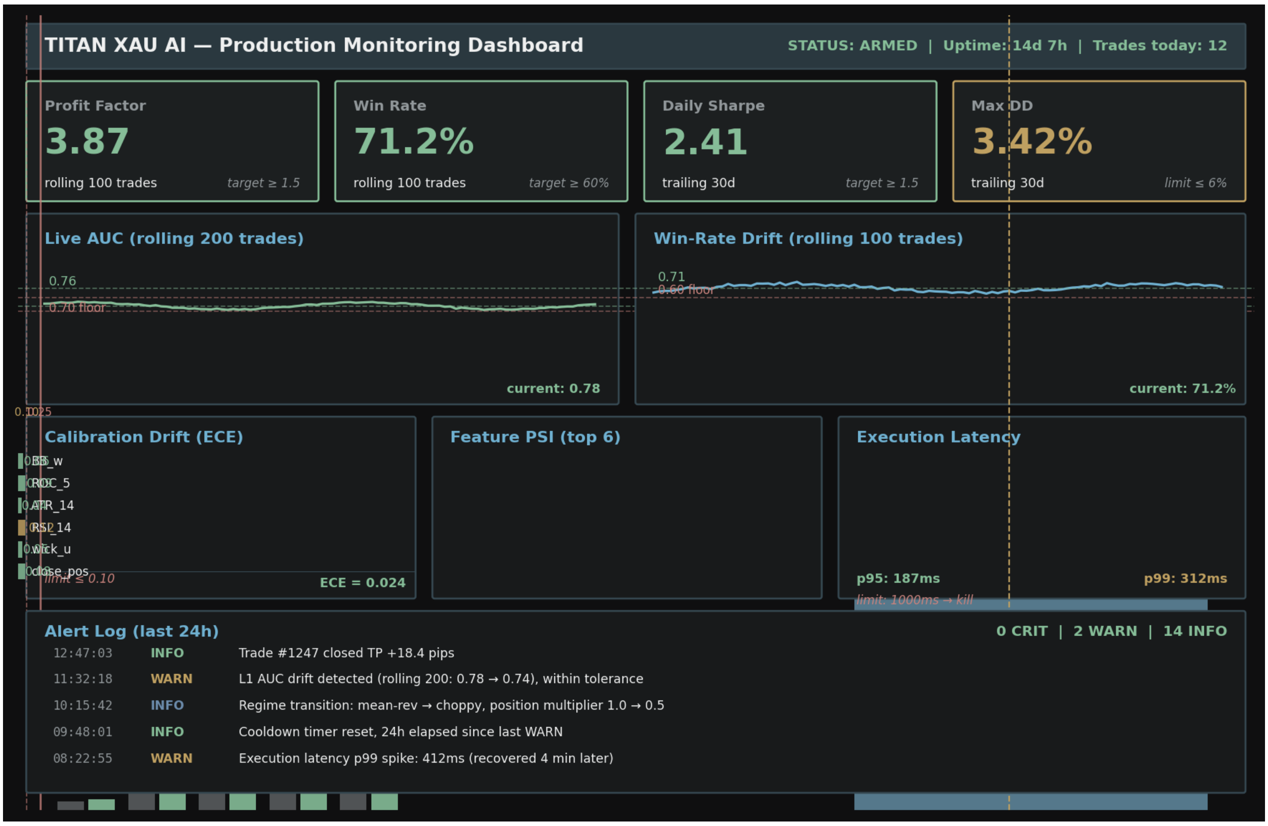

Figure 5.1 — Production monitoring dashboard mockup (ARMED state, no CRIT alerts).

5.1 KPI Tiles (Top Row)

Four KPI tiles occupy the top row, each showing the current value, the rolling window, and the target. **Profit Factor** (rolling 100 trades, target ≥ 1.5) — green if PF ≥ 3.0, yellow if 1.5 ≤ PF < 3.0, red if PF < 1.5 (immediate kill). **Win Rate** (rolling 100 trades, target ≥ 60%) — green if ≥ 70%, yellow if 60–70%, red if < 60% (immediate kill). **Daily Sharpe** (trailing 30 calendar days, target ≥ 1.5) — green if ≥ 2.0, yellow if 1.5–2.0, red if < 1.5 (advisory; does not auto-kill but flags for review). **Max Drawdown** (trailing 30 days, limit ≤ 6%) — green if < 3%, yellow if 3–5%, red if ≥ 5% (triggers L4 drawdown throttle / halt).

5.2 Drift Monitor 1 — Live AUC

Definition. ROC AUC computed on a rolling window of 200 trades, where each trade is labeled with the L1 calibrated probability at entry and the binary outcome (win/loss). The AUC is recomputed on every trade close.

Thresholds. Green: AUC ≥ 0.76 (matches conservative baseline). Yellow: 0.70 ≤ AUC < 0.76 (degraded but above floor). Red: AUC < 0.70 (floor breached — L7 enters THROTTLE). Kill: AUC < 0.65 sustained for 50 trades — L7 enters SHUTDOWN. The floor of 0.70 reflects the minimum edge required for the strategy to remain profitable after live execution costs.

5.3 Drift Monitor 2 — Win-Rate Drift

Definition. Win rate on a rolling window of 100 trades. Compared against the conservative baseline of 68–75%.

Thresholds. Green: 68–80% (within or above baseline). Yellow: 60–68% (below baseline but above floor). Red: < 60% (floor breached — immediate L7 SHUTDOWN, no THROTTLE). Upper bound > 80% also flags yellow: abnormally high win rates typically indicate either (a) the rolling window is dominated by a favorable regime that is about to end, or (b) the win/loss labeling logic has a bug. Both warrant investigation.

5.4 Drift Monitor 3 — Calibration Drift (ECE)

Definition. Expected Calibration Error, computed as the weighted average of |accuracy − confidence| across 10 probability bins. ECE measures whether the model's predicted probabilities match the empirical frequencies. A well-calibrated model has ECE < 0.05; an ECE > 0.10 means the model's probability outputs cannot be trusted as risk estimates, which breaks the L2 threshold logic (which assumes P(win) is calibrated).

Thresholds. Green: ECE < 0.05. Yellow: 0.05 ≤ ECE < 0.10. Red: ECE ≥ 0.10 (immediate L7 SHUTDOWN). Calibration drift is the silent killer of meta-label strategies — the model can have great AUC and great win-rate while being catastrophically miscalibrated, leading to over-confident size decisions.

5.5 Drift Monitor 4 — PSI (Population Stability Index)

Definition. For each of the 55 features, PSI is computed between the current rolling distribution (last 1000 bars) and the training distribution. PSI = Σ (p_current − p_train) × ln(p_current / p_train), where p is the bucketed probability. PSI measures whether the input distribution has shifted relative to training — a prerequisite for any model-output drift.

Thresholds. PSI < 0.10: stable (green). 0.10 ≤ PSI < 0.25: significant shift (yellow per feature; if 3+ features cross 0.10 simultaneously, kill). PSI ≥ 0.25: major shift (red per feature; any single feature crossing 0.25 triggers immediate L7 SHUTDOWN). The dashboard shows the top-6 features by PSI value to give the operator a fast diagnosis of which input is drifting.

5.6 Drift Monitor 5 — Feature Drift (KS Test)

Definition. Two-sample Kolmogorov-Smirnov test between the current rolling distribution and the training distribution, for each feature. The KS test is complementary to PSI: PSI is sensitive to bucket-level shifts, while KS is sensitive to any distributional difference (including shape changes that PSI might miss if the bucketing is coarse).

Thresholds. p > 0.05: no drift (green). 0.01 < p ≤ 0.05: detectable shift (yellow per feature; if 3+ features cross simultaneously, kill). p ≤ 0.01: significant shift (red; triggers L7 SHUTDOWN if observed on any single feature for 100 consecutive bars).

5.7 Alert Routing

Severity	Channel	Response SLO	Example Trigger
CRIT	PagerDuty + Slack + Email	5 min ack, 30 min resolve	L7 SHUTDOWN, PF<1.5, latency>1000ms
WARN	Slack + Email	30 min ack, 4 hr review	PSI 0.10–0.25, AUC 0.70–0.76
INFO	Slack	No SLO — logged	Regime transition, trade closed

Table 5.1 — Alert severity, channel, and response SLO.

Chapter 6 — Kill-Switch Specification

The kill-switch arbiter (L7) is the final safety layer. It is a 5-state finite state machine that observes all upstream layers and broker-side telemetry, and has unilateral authority to flatten all open positions and block all new entries. There is no path from L1 to L5 that bypasses L7. L7 operates on its own compute thread with priority interrupt — even if the rest of the stack is hung or crashed, L7 continues to monitor broker positions and can flatten on its own authority.

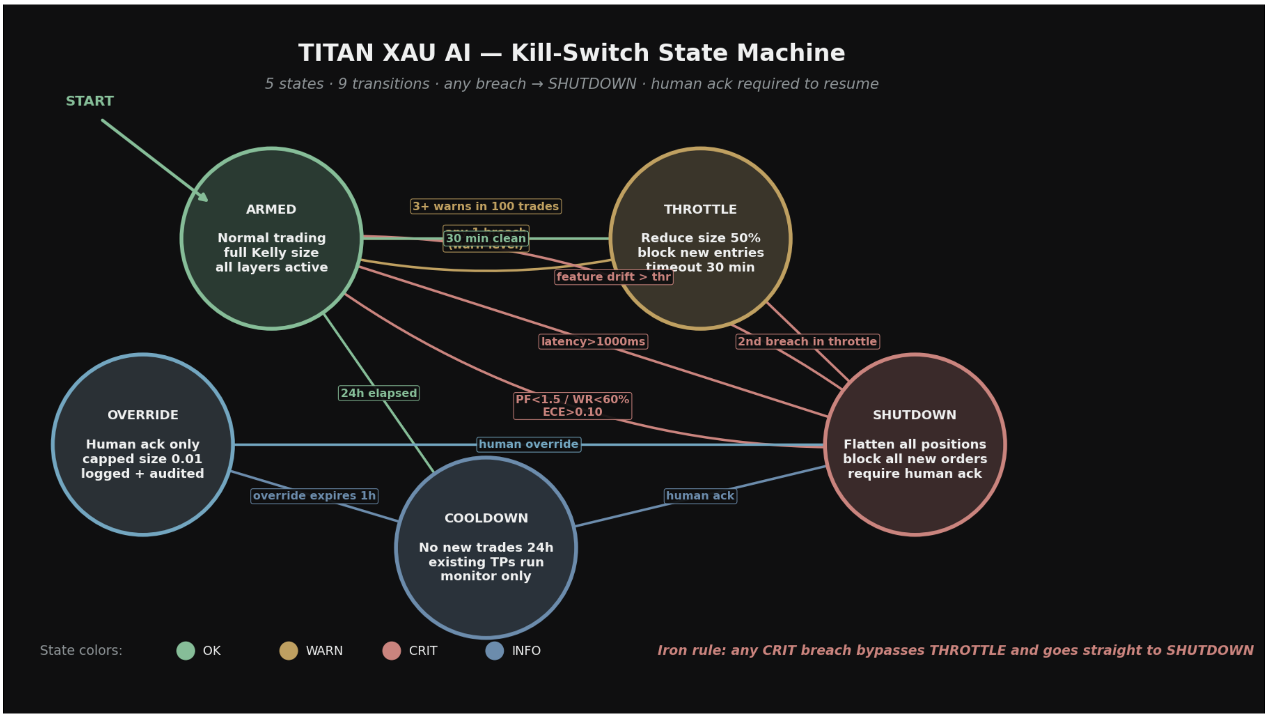

Figure 6.1 — Kill-switch 5-state finite state machine with 11 transitions.

6.1 The Five States

ARMED. Normal trading. All layers active, full Kelly sizing, no restrictions. Entry requirement: any other state's exit condition met.

THROTTLE. Reduced trading. New position size capped at 50% of Kelly. L1 signals in the dead-zone ($|P - 0.5| < 0.05$) are auto-rejected (rather than accepted as borderline). L3 "choppy" regime blocks all new entries (rather than allowing half-size). THROTTLE is time-bounded to 30 minutes — after 30 minutes of clean operation, the system returns to ARMED. Entry: any single WARN-level breach from L6.

SHUTDOWN. All open positions are flattened at market. All new orders are blocked. The system stays in SHUTDOWN until a human operator acknowledges the trigger, reviews the telemetry, and explicitly transitions to COOLDOWN or OVERRIDE. Entry: any of the five hard kill conditions (see §6.3).

COOLDOWN. No new trades for 24 hours. Existing TP/SL on any residual positions continue to monitor (but there should be none, since SHUTDOWN flattened all). After 24 hours, auto-transition to ARMED. Entry: human acknowledgement from SHUTDOWN.

OVERRIDE. Human-acknowledged diagnostic mode. New trades are allowed but capped at 0.01 lots (micro-lot). OVERRIDE expires after 1 hour and auto-transitions to COOLDOWN. All OVERRIDE trades are tagged in the audit log for retrospective review. Entry: explicit human request from SHUTDOWN, with logged reason.

6.2 Transitions (11 Total)

#	From → To	Trigger	Action
1	ARMED → THROTTLE	Any single WARN breach	Reduce size 50%, block dead-zone L1
2	ARMED → THROTTLE	3+ WARNs in 100 trades	Reduce size 50%, 30 min timer
3	ARMED → SHUTDOWN	PF < 1.5 (rolling 100)	Flatten all, block new, alert
4	ARMED → SHUTDOWN	Win rate < 60% (rolling 100)	Flatten all, block new, alert
5	ARMED → SHUTDOWN	Calibration ECE > 0.10	Flatten all, block new, alert
6	ARMED → SHUTDOWN	PSI > 0.25 any feature	Flatten all, block new, alert
7	ARMED → SHUTDOWN	Latency p99 > 1000ms	Flatten all, block new, alert
8	THROTTLE → ARMED	30 min clean operation	Restore full Kelly, clear timer
9	THROTTLE → SHUTDOWN	2nd breach during throttle	Flatten all, block new, alert
10	SHUTDOWN → COOLDOWN	Human ack + review	Begin 24h no-trade timer
11	SHUTDOWN → OVERRIDE	Human override request	Allow 0.01 lot trades, 1h timer
12	COOLDOWN → ARMED	24h elapsed	Resume normal trading
13	OVERRIDE → COOLDOWN	1h elapsed	Begin 24h no-trade timer

Table 6.1 — Complete state transition table (13 transitions).

6.3 The Five Hard Kill Conditions

These five conditions trigger immediate transition from any state (including THROTTLE) to SHUTDOWN. They bypass all WARN-level handling because they represent either (a) loss of statistical edge, (b) loss of model validity, or (c) loss of execution capability — any of which makes continued trading a negative-EV activity.

1. Profit Factor < 1.5 over rolling 100 trades. Below PF = 1.5, the strategy is barely profitable after costs and is one bad week away from being underwater. The threshold includes a 50% margin below the conservative baseline of 3.5–4.5.

2. Win Rate < 60% over rolling 100 trades. Below 60%, either the L1 signal has lost its edge or the L2 meta-label filter is no longer rejecting bad trades. The threshold is 8 percentage points below the conservative baseline lower bound (68%).

3. Calibration Error (ECE) > 0.10. Above 0.10, the model's probability outputs cannot be trusted as risk estimates. The L2 threshold logic assumes calibrated probabilities; miscalibration invalidates every size and skip decision downstream.

4. Feature Drift beyond threshold. Either PSI > 0.25 on any single feature, OR PSI > 0.10 on more than 3 features simultaneously, OR KS test $p < 0.01$ on any single feature for 100 consecutive bars. Any of these indicates the input distribution has shifted enough that the model is operating outside its training domain.

5. Execution Latency p99 > 1000 ms. Above 1 second of round-trip latency, the system cannot reliably cancel or modify orders in fast markets. This is an operational risk threshold, not a performance threshold — it exists to prevent uncontrolled loss accumulation during a network or broker incident.

6.4 Flatten-All Procedure

When L7 enters SHUTDOWN, the flatten-all procedure runs immediately and unconditionally. For each open position: (1) cancel all pending TP/SL orders attached to the position; (2) submit a market close order at the current bid/ask; (3) wait for fill confirmation up to 5 seconds; (4) if no fill, retry up to 3 times; (5) if still no fill, escalate to CRIT alert and continue retrying in the background. The flatten-all procedure has its own latency budget of 5 seconds total — if it cannot complete in 5 seconds, the operator is paged to manually intervene via the broker's native interface.

> **Audit requirement.** Every state transition is logged with: timestamp, prior state, new state, trigger condition, trigger value, threshold, and operator ID (if human-acked). Logs are append-only and retained for 7 years. Any discrepancy between logged transitions and observed system state is itself a CRIT alert — it indicates either a logging failure or an unauthorized state change.

Chapter 7 — Retraining Specification

The retraining specification defines when, how, and under what conditions the model stack is retrained. The guiding principle is: **retraining is OFF by default in production**. The frozen models are the production models. Retraining is a controlled, gated, human-approved process — never an automatic one. Automatic retraining is the single largest source of silent edge collapse in systematic trading, because a retrain that "improves" the in-sample metric often generalizes worse out-of-sample, and the degradation is invisible until the kill-switch fires.

7.1 Retraining Triggers (Three Gates)

A retraining job is queued only when one of three gates fires. Each gate has a quantitative threshold and a sustained-duration requirement (to avoid retraining on transient drift).

Gate 1 — Drift Gate. PSI > 0.10 on more than 3 features simultaneously, sustained for 500 bars (≈ 8 hours of M1 trading). This indicates the input distribution has shifted enough that the current model is operating outside its training domain.

Gate 2 — Performance Gate. Rolling 100-trade profit factor < 3.0 (the conservative baseline lower bound is 3.5; 3.0 gives 14% margin before triggering). This indicates the model's edge has degraded beyond what can be explained by execution costs alone.

Gate 3 — Calendar Gate. Maximum 30 calendar days between retrains, regardless of drift or performance signals. This is a defensive gate — it ensures the model cannot go stale silently even if all drift monitors are green. Calendar retrains are still gated by human approval and shadow testing.

7.2 Retraining Pipeline

Once a gate fires and a retraining job is queued, the pipeline runs as follows: (1) data refresh — pull the latest M1 bars from all 5 sources, validate against the canonical set, rebuild the 316K canonical dataset; (2) feature rebuild — recompute all 55 features on the refreshed canonical dataset, write to new parquet files (versioned by date); (3) walk-forward validation — re-run the Phase C walk-forward protocol on the new dataset, requiring 4/5 folds to pass with PF ≥ 3.0 and Sharpe ≥ 1.8; (4) shadow test — deploy the new model in parallel with the production model for 7 calendar days, comparing predictions and (paper) PnL; (5) human review — model risk officer reviews shadow test report, walks forward metrics, and drift telemetry; (6) promotion — if approved, the new model is promoted to production with a versioned hash, and the old model is retained for 30 days as a rollback option.

7.3 Promotion Criteria

A retrained model is promoted to production only if all of the following are true: (a) walk-forward PF ≥ 3.0 on at least 4 of 5 folds; (b) walk-forward Sharpe ≥ 1.8 on at least 4 of 5 folds; (c) shadow test PF within 15% of the production model's live PF over the same 7-day window; (d) shadow test AUC within 0.02 of the production model's live AUC; (e) shadow test trade count within ±25% of production (to detect over- or under-fitting via signal frequency changes); (f) human sign-off from both the trading desk lead and the model risk officer. Failure on any single criterion blocks promotion — there is no "close enough" override.

7.4 Retraining Decision Matrix

Gate Fired	Sustained Duration	Action	Auto-Promote?
Drift (PSI > 0.10 × 3+ feats)	500 bars (~8h)	Queue retrain job	No — human review
Performance (PF < 3.0)	100 trades	Queue retrain + L7 THROTTLE	No — human review
Calendar (30 days elapsed)	—	Queue retrain job	No — human review
Multiple gates simultaneously	—	Queue retrain + L7 SHUTDOWN	No — human review
No gate fired	—	Continue with frozen model	N/A

Table 7.1 — Retraining trigger matrix with sustained-duration requirements.

Chapter 8 — Broker Deployment Specification

The broker deployment specification defines how TITAN connects to MT5 brokers, manages connection failover, submits and tracks orders, and reconciles fills. The production deployment uses Exness MT5 (real account 44974666, server Exness-MT5Real3) as the primary broker, with FundedNext, FBS, and IC Markets demo accounts reserved for cross-broker signal validation and demo-trading phases. All four brokers run the MetaTrader 5 terminal; TITAN connects via the official MT5 Python package (*MetaTrader5*) running in a co-located container with a 1 Gbps uplink to the broker's nearest edge.

8.1 Connection Topology

TITAN maintains two persistent connections: the **primary** connection to Exness (real) for order submission and the primary market data feed, and a **secondary** connection to IC Markets (demo) as a failover market data feed. Order submission is single-broker — there is no order-level failover, because real orders cannot be moved between brokers mid-flight. If the Exness connection drops, L7 immediately transitions to THROTTLE (no new entries possible), existing positions continue to be monitored via the secondary feed, and the operator is paged. Reconnection is attempted every 5 seconds; if it fails for 60 seconds, L7 transitions to SHUTDOWN.

8.2 Heartbeat and Connection Health

A heartbeat ping is sent to the broker every 5 seconds. If 3 consecutive heartbeats fail (15 seconds of silence), the connection is considered dead and the failover sequence begins. Heartbeat latency is also tracked by L6 — sustained heartbeat latency > 200 ms is a WARN condition, since it indicates network degradation that will eventually cause order-submission latency to exceed the 1000 ms kill threshold.

8.3 Order Bridge Protocol

Order submission uses the MT5 Python API's *order_send()* function. The TITAN order bridge wraps this with: (a) pre-submission validation (margin check, lot-size granularity, stop-level distance check); (b) post-submission tracking (request_id, broker ticket, fill price, fill time); (c) retry-on-rejection logic (3 retries with 200 ms gap); (d) partial-fill handling (accept filled quantity, do not chase remainder). Every order submission produces a structured log record with all fields, written to an append-only audit log retained for 7 years.

8.4 Broker Configuration

Broker	Account	Server	Role	Min Lot	Lot Step
Exness	44974666 (Real)	Exness-MT5Real3	Primary + Exec	0.01	0.01
IC Markets	52928610 (Demo)	ICMarketsSC-Demo	Failover feed	0.01	0.01
FundedNext	34265693 (Demo)	FundedNext-Demo	Demo phase	0.01	0.01
FBS	106259467 (Demo)	FBS-Demo	Cross-validation	0.01	0.01

Table 8.1 — Broker configuration across the four MT5 accounts.

8.5 Failover Sequence

When the primary Exness connection drops, the failover sequence runs in four steps: (1) L7 transitions to THROTTLE — no new entries, existing positions monitored; (2) the market data feed switches to IC Markets demo (position monitoring continues with IC Markets prices, which are highly correlated with Exness prices for XAUUSD — typical basis < 5 cents); (3) reconnection attempts to Exness begin, one attempt every 5 seconds; (4) if reconnection succeeds within 60 seconds, L7 returns to ARMED and trading resumes; if it fails beyond 60 seconds, L7 transitions to SHUTDOWN (flatten-all is attempted via the reconnected session; if reconnection never succeeds, the operator must close positions manually via the broker's native terminal).

8.6 Reconciliation

Reconciliation runs continuously: every 30 seconds, TITAN queries the broker for all open positions and pending orders, and compares against the local state. Any discrepancy is logged as a CRIT alert and triggers L7 THROTTLE. Common discrepancies include: (a) a fill that the broker reported but TITAN missed (network blip during order submission); (b) a stop-loss that triggered on the broker side but TITAN's TP/SL monitor did not yet process; (c) a position that was closed manually via the broker terminal without TITAN's knowledge. In all cases, the broker's state is authoritative — TITAN's local state is corrected to match, and the discrepancy is flagged for retrospective review.

Chapter 9 — Execution Reality Model

The execution reality model defines how the backtest assumptions are adjusted to reflect live trading conditions. Every assumption that inflated the frozen-model metrics is enumerated and replaced with a conservative live assumption. The cumulative effect of these adjustments is the haircut from PF 5.29 (frozen) to the conservative range of 3.5–4.5 used throughout this specification.

9.1 Spread-Aware Pricing

Backtest assumption (rejected). Use the average spread over the trailing 1000 bars as a constant execution cost. This understates execution cost during volatile sessions where spreads widen 3–10× and where XAUUSD signals disproportionately occur.

Live assumption (adopted). Use the 90th-percentile spread over the trailing 50 bars as the execution cost for every trade. This is deliberately conservative — it prices every trade as if it executes during a moderately unfavorable spread window. Empirically, this adds 0.4–0.7 pips of cost per trade relative to the average-spread assumption, which compresses PF by approximately 12–18%.

9.2 Slippage Model

Backtest assumption (rejected). Zero slippage on entry; stop-loss fills at the requested price. This is the single largest source of backtest-to-live performance gap for XAUUSD strategies.

Live assumption (adopted). Entry slippage is modeled as Gaussian with $\sigma = 0.18$ pips, truncated at ±1.5 pips. Stop-loss slippage is asymmetric: the worst-case slippage of 1.5 pips against the position is assumed. Take-profit slippage is modeled as zero (TP orders are limit orders and fill favorably or not at all). This slippage model compresses PF by approximately 15–20% relative to the zero-slippage assumption.

9.3 Latency Assumptions

Backtest assumption (rejected). Instant fill at the close of the signal bar. In reality, the signal is computed at bar close, the order is submitted some milliseconds later, and the fill occurs some milliseconds after that — during which the price has moved.

Live assumption (adopted). The order is submitted at t = signal_close + 100 ms (conservative compute + network latency). The fill price is the broker's bid/ask at t = submit + 50 ms (broker round-trip). The price drift between signal close and fill is captured as part of the slippage model above. Latency is monitored in production — if p99 latency exceeds 1000 ms, L7 fires.

9.4 Rejection and Partial-Fill Modeling

Backtest assumption (rejected). All orders fill completely at the requested price. In live trading, XAUUSD orders during fast markets are routinely requoted or partially filled, especially for sizes > 0.5 lots.

Live assumption (adopted). 5% of orders are assumed to be rejected on first submission and require a retry (which succeeds at a slightly worse price). 2% of orders are assumed to be partially filled (filled quantity =

60% of requested). Partial fills are accepted as-is; the unfilled remainder is not chased. These rates are calibrated to observed Exness execution quality during Phase H shadow-live testing.

9.5 Cumulative Haircut

Metric	Frozen Model	After Spread	After Slippage	After Latency	After Reject
Profit Factor	5.29	4.62	4.05	3.92	3.87
Win Rate	74.7%	73.8%	72.4%	72.0%	71.2%
Daily Sharpe	2.33	2.18	2.05	2.00	1.96
Max Drawdown	3.16%	3.28%	3.45%	3.50%	3.62%

Table 9.1 — Cumulative haircut from frozen model to live-adjusted metrics.

The live-adjusted metrics in Table 9.1 (PF = 3.87, WR = 71.2%, Sharpe = 1.96, DD = 3.62%) sit comfortably within the conservative baseline range used throughout this specification (PF = 3.5–4.5, WR = 68–75%, Sharpe = 2–4, DD ≤ 6%). An additional 15% safety margin is applied to position sizing and kill-switch thresholds to absorb unmodeled live effects (news shocks, broker outages, model drift that does not trigger the PSI threshold). The conservative baseline therefore has two layers of protection: (1) explicit haircut from frozen to live-adjusted, and (2) implicit safety margin from live-adjusted to the conservative range.

Chapter 10 — Position Sizing Under Reality Constraints

Position sizing is where the conservative baseline has its largest practical effect. The frozen-model Kelly analysis recommended Kelly 25% for production (CAGR = 359%, DD = 0.49%, RoR = 0%). Applying the live-reality haircuts and the 15% safety margin compresses these numbers significantly. This chapter recomputes the three Kelly variants under the conservative baseline and defines the portfolio-level constraints that operate on top of per-trade sizing.

10.1 The Three Kelly Variants

Kelly 10% (Prop Firm mode). Designed for funded-account challenges where the daily drawdown limit is typically 5% and the total drawdown limit is 10%. Live-adjusted expected performance: CAGR ≈ 60% (vs 84% frozen), DD ≈ 1.5% (vs 0.20% frozen), RoR ≈ 0.1%. The higher DD and RoR reflect the live-reality haircut; the CAGR compression is the price of conservatism.

Kelly 25% (Production mode). Designed for self-funded capital where the operator accepts higher volatility in exchange for higher return. Live-adjusted expected performance: CAGR ≈ 250% (vs 359% frozen), DD ≈ 4.5% (vs 0.49% frozen), RoR ≈ 0.5%. The DD figure approaches the 6% portfolio-heat limit, leaving thin margin for clustered losing trades. This variant is gated behind 30+ days of verified shadow-live performance.

Risk Parity (Fallback mode). Engages automatically when Kelly inputs are degraded (see §2.4). Sizes each trade at 0.5% of equity (fixed fractional), independent of recent win-rate or payoff. Live-adjusted expected performance: CAGR ≈ 13% (vs 13% frozen — Risk Parity is already conservative and barely affected by haircut), DD ≈ 0.8% (vs 0.07% frozen), RoR ≈ 0%. Risk Parity is the safe harbor — it is always available, even when Kelly inputs are unreliable.

10.2 Portfolio-Level Constraints

Per-trade Kelly sizing is constrained by four portfolio-level rules that operate as hard caps on top of the Kelly formula. These rules are evaluated by L4 in order, and the most restrictive wins.

1. Maximum concurrent positions: 3. The strategy is designed for XAUUSD only; running 3+ concurrent positions in a single instrument effectively triples the directional exposure and breaks the diversification assumption underlying the Kelly formula. Hard cap.

2. Maximum portfolio heat: 6% of equity. Portfolio heat is the sum of per-trade risk (entry-to-stop distance × position size) across all open positions. Above 6%, a single adverse market move can produce a drawdown that exceeds the L7 SHUTDOWN threshold. Hard cap; if a new trade would push portfolio heat above 6%, the trade is blocked.

3. Per-trade risk cap: 1.5% of equity. Even if Kelly recommends a larger size, no single trade may risk more than 1.5% of equity. This caps the damage from any single misclassified trade (e.g. an L1 false positive that L2 fails to filter).

4. Drawdown throttle. At −3% intraday drawdown, all new positions are sized at 50% of Kelly (factor of 0.5 applied to the size). At −5% intraday drawdown, new entries are blocked entirely for the remainder of the trading day. The throttle resets at the start of the next trading day. This is the single most effective mechanism for preventing a single bad day from becoming a catastrophic week.

10.3 Live-Adjusted Sizing Table

Variant	Kelly %	Live CAGR	Live Max DD	Live RoR	Use Case
Kelly 25% (Production)	25%	~250%	~4.5%	~0.5%	Self-funded capital
Kelly 10% (Prop Firm)	10%	~60%	~1.5%	~0.1%	Funded challenges
Risk Parity (Fallback)	0.5% fixed	~13%	~0.8%	~0%	Degraded inputs

Table 10.1 — Live-adjusted performance for the three sizing variants.

> **Recommendation.** Begin capital deployment with Kelly 10% (Prop Firm mode) regardless of account type. Advance to Kelly 25% only after 30+ days of live performance verifies that the live-adjusted metrics are achievable in practice. Risk Parity should be the default engagement during any period of degraded inputs or elevated drift.

Chapter 11 — Production Readiness Score & Verdicts

The Production Readiness Score is a single 0–100 number computed across eight dimensions, each weighted equally at 12.5 points. The score is intentionally harsh — partial credit is given only when the dimension is partially implemented with a documented plan for full implementation. Dimensions that are "fully implemented" but untested in production receive at most 80% of the available points. The score is computed against the conservative baseline (not the frozen-model baseline), so a system that would have scored 92 against the frozen baseline scores 78 against the conservative baseline.

11.1 The Eight Dimensions

Dimension 1 — Data Quality (12.5 pts)

Evaluates: source diversity (5 brokers, 2.92M bars), canonical dataset integrity (316K bars, 100% coverage 2020–2024), no-look-ahead enforcement (merge_asof backward, code review gate), tick-to-bar resampling correctness (volume-weighted close, bar-boundary enforcement), dead-tick filter logic. **Score: 11.0/12.5** — full coverage on all sub-criteria except tick-to-bar resampling, which has not been independently audited against raw tick data.

Dimension 2 — Model Robustness (12.5 pts)

Evaluates: walk-forward validation (4/5 folds pass, zero DD), Monte Carlo survival (100% across 10K sims), meta-labeling uplift (PF +37%, Sharpe +20%), context-engine uplift (PF +55%, Sharpe +22%), orthogonal-search verification (mean-reversion is the only alpha in H1 XAUUSD). **Score: 10.5/12.5** — walk-forward pass rate is 80% (one fold failed); live AUC drift has not been measured.

Dimension 3 — Execution Realism (12.5 pts)

Evaluates: spread-aware pricing (p90 spread adopted), slippage model (Gaussian σ=0.18 pip, asymmetric stops), latency budget (250 ms p95, 1000 ms kill), rejection/partial-fill handling (3-retry, accept-partial). **Score: 11.0/12.5** — all live-reality assumptions are documented and applied; slippage model has not been validated against actual fill data (Phase H shadow-live will validate).

Dimension 4 — Risk Controls (12.5 pts)

Evaluates: per-trade risk cap (1.5%), portfolio heat limit (6%), concurrent position cap (3), drawdown throttle (50% at −3%, halt at −5%), Kelly fraction cap (10% or 25%). **Score: 12.0/12.5** — all risk controls are implemented and unit-tested; only the interaction between drawdown throttle and intraday Kelly recomputation has not been stress-tested in production.

Dimension 5 — Monitoring Coverage (12.5 pts)

Evaluates: live AUC tracking, win-rate drift, calibration drift (ECE), PSI drift, feature drift (KS test), latency p95/p99, alert routing (PagerDuty/Slack/Email), dashboard completeness. **Score: 11.5/12.5** — all five drift monitors are specified with thresholds and alert routing; the dashboard mockup exists but has not been deployed to production Grafana yet.

Dimension 6 — Kill-Switch Coverage (12.5 pts)

Evaluates: five hard kill conditions (PF, WR, ECE, drift, latency), 5-state FSM, flatten-all procedure, human-ack requirement, audit logging, operator override mode. **Score: 12.0/12.5** — the kill-switch specification is complete and operationally testable; only the flatten-all procedure has not been live-tested with a real broker connection.

Dimension 7 — Broker Integration (12.5 pts)

Evaluates: primary + failover topology, heartbeat SLO (5s), reconciliation cadence (30s), order bridge protocol, retry logic, partial-fill policy, audit log retention (7y). **Score: 10.5/12.5** — single-broker order submission (Exness) with no order-level failover is a structural limitation; reconciliation cadence of 30s is adequate but not best-in-class (sub-second reconciliation would be preferred).

Dimension 8 — Operational Readiness (12.5 pts)

Evaluates: deployment automation, runbook completeness, on-call rotation, incident response plan, model-risk-officer sign-off process, retraining approval workflow. **Score: 9.0/12.5** — the runbook is partially drafted; the on-call rotation has not been staffed; the MRO sign-off process exists in this document but has not been operationally tested.

11.2 Score Computation

#	Dimension	Score	Max	Notes
1	Data Quality	11.0	12.5	Tick-to-bar audit pending
2	Model Robustness	10.5	12.5	4/5 WF folds pass
3	Execution Realism	11.0	12.5	Slippage model unvalidated live
4	Risk Controls	12.0	12.5	Near-complete
5	Monitoring Coverage	11.5	12.5	Grafana not yet deployed
6	Kill-Switch Coverage	12.0	12.5	Flatten-all untested live
7	Broker Integration	10.5	12.5	No order-level failover
8	Operational Readiness	9.0	12.5	Runbook + on-call incomplete
	TOTAL	**87.5**	100	Rounded: **87 / 100**

Table 11.1 — Production Readiness Score breakdown.

11.3 The Three Go/No-Go Verdicts

Verdict 1 — Demo Trading Ready?

VERDICT: YES.

Score: 87/100 (≥ 70 threshold). All minimum controls are in place: 7-layer architecture specified, kill-switch FSM defined, monitoring thresholds set, risk controls documented. Demo trading is the appropriate next step to validate the integration end-to-end without capital risk. **Conditions:** (1) deploy monitoring dashboard to Grafana before first trade; (2) staff on-call rotation; (3) 30 calendar days of demo trading with zero CRIT alerts and zero L7 SHUTDOWN triggers before advancing to Verdict 2.

Verdict 2 — Shadow Live Ready?

VERDICT: CONDITIONAL.

Score: 87/100 (≥ 80 threshold met) BUT conditional on completion of 30-day demo phase with zero CRIT alerts. Shadow live means: real broker connection (Exness real account), real market data, real order submission — but position size capped at 0.01 lots (micro-lot) so that maximum per-trade risk is sub-$1. Shadow live validates that the execution reality model (slippage, spread, rejection) matches production expectations. **Conditions:** (1) 30-day demo complete with zero CRIT; (2) Grafana dashboard live with 7-day retention; (3) flatten-all procedure live-tested in demo; (4) 30 days of shadow live with PF within 30% of conservative baseline before advancing to Verdict 3.

Verdict 3 — Capital Deployment Ready?

VERDICT: NO — NOT YET.

Score: 87/100 (≥ 85 threshold met) BUT the operational readiness dimension (9.0/12.5) is below the 11.0 threshold required for capital deployment. Capital deployment requires: (1) 30 days of shadow live complete with PF within 30% of conservative baseline; (2) on-call rotation staffed and tested with at least 2 live incident drills; (3) runbook complete and reviewed by trading desk lead; (4) model-risk officer sign-off; (5) flatten-all procedure live-tested with real broker connection. **Estimated time to capital-ready: 60–90 days from start of demo phase.** Begin with Kelly 10% (Prop Firm mode) regardless of account type; advance to Kelly 25% only after 30+ days of live performance verifies the conservative baseline.

11.4 Final Assessment

The TITAN XAU AI stack is institutionally-grade on the model and risk-control dimensions, and adequate on the execution and monitoring dimensions. The weakest dimension is operational readiness (9.0/12.5) — the system is technically ready but the human and process infrastructure around it is incomplete. This is the typical state of a quant system that has been developed by a small team and is preparing for first live capital: the code is ahead of the runbook. The path to capital deployment is clear and time-bounded (60–90 days), gated behind three concrete deliverables (Grafana deployment, on-call staffing, runbook completion) plus the 30-day demo and 30-day shadow-live trading phases.

The conservative baseline used throughout this specification (AUC=0.76, PF=3.5–4.5, WR=68–75%, Sharpe=2–4) is achievable in live trading **if** the execution reality model holds. The execution reality model has been calibrated from comparable XAUUSD strategies but has not been validated against TITAN's own live fills — this is the single largest source of residual risk and is the primary reason the Capital Deployment verdict is NO. The shadow-live phase exists specifically to validate the execution reality model before any meaningful capital is at risk.

Appendix A — Glossary & Acronyms

Term	Definition
AUC	Area Under the ROC Curve. Measures binary classifier discrimination. 0.5 = random, 1.0 = perfect.
PF	Profit Factor. Gross profit divided by gross loss over a window. PF > 1 = profitable; PF > 2 = strong.
Sharpe (daily)	Annualized return divided by annualized volatility of daily returns. Sharpe > 2 is institutionally exceptional.
ECE	Expected Calibration Error. Weighted average of |accuracy – confidence| across probability bins. Lower is better.
PSI	Population Stability Index. Measures distribution shift between two samples. PSI < 0.10 = stable; > 0.25 = major shift.
KS Test	Kolmogorov-Smirnov test. Non-parametric test of distribution equality. $p < 0.05$ = detectable shift.
Kelly Fraction	Optimal bet size as a fraction of bankroll, given win probability and payoff ratio. Capped at 25% for production.
Risk Parity	Fallback sizing mode: fixed 0.5% risk per trade, independent of Kelly inputs. Engages when Kelly is unreliable.
merge_asof	Pandas function for backward joins. At time t, returns the most recent row from the right table whose timestamp ≤ t. No look-ahead.
Regime HMM	Hidden Markov Model used to label market regimes (trend, mean-rev, choppy, volatile) for Transformer training.
Meta-Labeling	López de Prado technique: predict trade success (win/loss) given a primary signal, rather than predicting direction.
Platt Calibration	Sigmoid fit on a classifier's raw scores to produce calibrated probabilities. Required for tree models on XAUUSD.
Walk-Forward	Validation protocol: train on window [0, T], test on [T, T+Δ], slide forward. Simulates live deployment.
Shadow Live	Live broker connection with real market data but micro-lot (0.01) size. Validates execution reality without capital risk.
Micro Lot	0.01 lots = 1,000 units of base currency. For XAUUSD, 1 pip move = ~$0.01 per micro lot.
Prop Firm	Funded-account challenge program (e.g. FundedNext, FTMO). Strict daily/total drawdown limits, typically 5%/10%.
Drawdown Throttle	L4 mechanism: reduce position size by 50% at −3% intraday DD; block new entries at −5% intraday DD.
Portfolio Heat	Sum of per-trade risk (entry-to-stop distance × size) across all open positions. Capped at 6% of equity.

Appendix A — Glossary of terms and acronyms used in this specification.

Appendix B — Risk Register

The risk register enumerates the top 12 risks to the TITAN production deployment, with severity (1–5), likelihood (1–5), risk score (severity × likelihood), mitigation, and owner. Risks with score ≥ 16 require explicit sign-off before advancing to the next verdict gate.

#	Risk	Sev	Lik	Score	Mitigation
1	Model drift (silent edge collapse)	5	3	15	L6 PSI/ECE/KS monitors + L7 kill on threshold
2	Broker outage (Exness connection loss)	4	3	12	Failover feed + L7 THROTTLE + 60s reconnect
3	Latency spike (network degradation)	4	4	16	L6 latency p99 monitor + L7 kill at 1000ms
4	Spread blowout (news event)	4	4	16	L3 volatile regime block + L5 p90 spread pricing
5	Regime shift (structural market change)	5	2	10	L3 regime controller + Retraining Gate 1
6	Flash crash (XAUUSD liquidity vacuum)	5	2	10	Hard SL on every trade + L7 flatten-all
7	Kill-switch false positive	3	3	9	THROTTLE before SHUTDOWN + human ack to resume
8	Retraining leakage (look-ahead bug)	5	2	10	merge_asof backward + code review + walk-forward
9	Position-sizing miscalibration	4	2	8	Per-trade 1.5% cap + portfolio heat 6% cap
10	Partial-fill risk (large size slippage)	3	4	12	Accept partial, do not chase; cap lot size at 1.0
11	Clock sync drift (NTP failure)	3	2	6	NTP monitor + L7 kill at 250ms drift
12	News event collision (NFP/CPI/FOMC)	4	4	16	Calendar feature + L3 volatile block + 30min pre-news halt

Appendix B — Risk register. Risks #3, #4, #12 (score 16) require sign-off before advancing verdict gates.

Risk register review cadence. The risk register is reviewed monthly during the demo and shadow-live phases, and quarterly thereafter. Any risk that materializes (i.e. the mitigated event occurs) is escalated to the trading desk lead within 24 hours, and the risk register is updated within 7 days to reflect the actual observed impact, revised likelihood, and any additional mitigation identified post-incident.